PROJECT BLUEPRINT: ALPHA-ML-RESEARCHER

Autonomous Quantitative Signal & Closed-Loop Control Systems Engine (Crypto Vault)

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1. Executive Overview & Mission

The **Alpha ML Researcher** is an industrial-grade trading intelligence engine fusing **Kaggle-grade Machine Learning** with **Classical & Modern Control Systems Engineering (ECE)**. It replaces heuristic indicators with deterministic state estimation, stationarity transformations, and closed-loop position sizing.

2. The 7 Constitutional Golden Rules

- **Rule 1: Pure Math & ECE Control Foundations:** Market microstructure, state estimation, and capital sizing are governed strictly by deterministic control theory and rigorous statistics. Zero arbitrary indicator magic numbers.
- **Rule 2: Reuse Verified Math Core:** Directly import and interface verified LTI representations (StateSpace, PID, LQR, stability analysis, root locus, Kalman) from the tested control_system project.
- **Rule 3: PyTorch Deep Learning Integration:** Deploy PyTorch temporal architectures (TCN / PatchTST) trained exclusively via Purged & Embargoed Cross-Validation to prevent lookahead bias.
- **Rule 4: Strict Discrete-Time (z-domain) Math:** All execution and filtering operate in discrete time. All closed-loop system poles must lie strictly inside the unit circle ($|z| < 1$).
- **Rule 5: Actuator Saturation & Anti-Windup Clamping:** Position size and leverage represent physical actuators. The integral term halts accumulation immediately upon hitting risk allocation caps.
- **Rule 6: Transport Delay & Latency Margins:** Explicitly model network and exchange latency (z^{-d}). Maintain a Phase Margin > 35 deg to prevent whipsaws and oscillatory instability.
- **Rule 7: Fractional Differentiation Stationarity Gate:** Raw prices are prohibited from entering ML models. Input data must be transformed with the minimum fractional order d^* passing the ADF stationarity test ($p < 0.05$).

3. The 4 Production-Grade Upgrades

- **--anti-windup:** Clamps the integral term (int e dt) to prevent explosive leverage growth during prolonged sideways/consolidation regimes.
- **--volatility-adaptive-gains:** Dynamically schedules Kp, Ki, Kd and LQR penalty matrices (Q, R) based on Garman-Klass / ATR volatility to damp aggressive sizing during liquidation wicks.
- **--meta-labeling:** Uses the Triple-Barrier Method. Decouples direction prediction (Up/Down) from bet sizing confidence (0.0 to 1.0) to filter false breakouts.
- **--zero-leakage-cv:** Enforces strict walk-forward cross-validation with purging and embargo buffers, guaranteeing zero future data leakage in training.

4. End-to-End Architectural Dataflow

Stage	Module / Technology	Primary Function
1. Live Ingestion	ccxt (REST/WebSockets)	Extracts 15m OHLCV, 20-level orderbook, funding rates.
2. State Denoising	Discrete Kalman Filter	Strips microstructure bounce; extracts latent fair price & velocity.
3. Feature Transform	FracDiff (d ≈ 0.38)	Preserves trend memory while ensuring ADF stationarity ($p < 0.05$).
4. Alpha Inference	PyTorch TCN / Meta-Labeling	Outputs directional probability & bet sizing confidence score.
5. Sizing & Risk	Closed-Loop PID / LQR Engine	Scales position via conviction, damps via D-term, guards phase margin.
6. Action Output	Trading Action Plan Brief	Recommends exact position size, entry bracket, and dynamic de-risk rule.

5. User Action Checklist (What We Need From You)

To bring this complete autonomous pipeline online, here is the exact checklist of items you need to configure or provide:

- **[ ] Google Workspace / Drive Authorization:** Authenticate the Google Workspace MCP (drive.search, drive.downloadFile, etc.) in your Antigravity environment so I can upload and sync documents directly into your Drive folders.
- **[ ] SQLite Vault Placement:** Confirm where local historical market tick/candle databases should live (e.g., C:/Users/gotog/quant_data/crypto_vault.db).
- **[ ] Kaggle API Credentials (Optional GPU Compute):** Ensure kaggle.json is present in ~/.kaggle/ if you want the agent to offload heavy multi-epoch PyTorch training jobs directly to free cloud GPUs.
- **[ ] Exchange Target Selection:** Confirm your primary target pairs (e.g., BTC/USDT, ETH/USDT on Binance or Bybit public endpoints). Zero exchange API keys or account balances are required for data fetching.
- **[ ] Risk Boundaries Configuration:** Define your account safety parameters: Maximum leverage cap (e.g., 3x), maximum single-position equity risk (e.g., 2% of portfolio), and daily drawdown cutoff.

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